

Question	Marking Guidance	Mark	Comments
03.1	1. Mutation in the viral DNA/RNA/genome/genetic material; 2. Altered (tertiary structure of the) viral attachment protein; 3. Allows it/attachment protein/virus to bind (to receptors of other species);	2 max	1. Accept named examples mutations 2. Accept 'antigen' for 'attachment protein' 2. Accept causes antigenic variability 3. Accept descriptions of binding eg is complementary
03.2	For one mark, accept any two of the following: • The polymerase chain reaction • Genetic/DNA fingerprinting • (Gel) electrophoresis • DNA/genome sequencing;	1	Accept PCR for polymerase chain reaction Accept autoradiography Accept DNA hybridisation Accept compare DNA/base sequence for 'DNA sequencing' Ignore compare mRNA base sequence Ignore compare amino acid sequence Ignore DNA probes
03.3	1. (The scientists) could identify proteins (that derive from the genetic code) OR (The scientists) could identify the proteome; 2. (They) could (then) identify potential antigens (to use in the vaccine);	2	2. Reject if answer suggests vaccine contains antibodies

03.4	1. B cell (antibody) binds to (viral) specific/complementary receptor/antigen; 2. B cell clones OR 3. <u>Plasma cells</u> release/produce (monoclonal) <u>antibodies</u> (against the virus); 4. (B/plasma cells produce/develop) memory cells;	3 max	1. Accept <u>B cell</u> forms antigen-antibody complex 2. Accept B cell undergoes clonal selection/expansion
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